# Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 03

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|---|---|
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| Specialization | Software Engineering |
| Batch No | 1 |

Activity 1

Create a new C program file and type the below program. Compile and execute it. Observe the result and explain it in your words

```
kamshiga@kamshiga-VirtualBox: ~
  GNU nano 7.2                                    ta1.c
#include <stdio.h>
int main()
{
 int i=1;

 while(i <= 5){
  printf("Value of i is: %d\n",i);
  i++;
 }

 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ta1.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta1 ta1.c
kamshiga@kamshiga-VirtualBox:~$ ./ta1
Value of i is: 1
Value of i is: 2
Value of i is: 3
Value of i is: 4
Value of i is: 5
kamshiga@kamshiga-VirtualBox:~$
```

Activity 2

Rewrite the program given in above Activity 1 using a for loop.

```
kamshiga@kamshiga-VirtualBox: ~
  GNU nano 7.2                                    ta2.c
#include <stdio.h>
int main()
{
 for(int i=1;i<=5;i++){
  printf("Value of i is: %d\n", i);
 }

 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ta2.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta2 ta2.c
kamshiga@kamshiga-VirtualBox:~$ ./ta2
Value of i is: 1
Value of i is: 2
Value of i is: 3
Value of i is: 4
Value of i is: 5
kamshiga@kamshiga-VirtualBox:~$
```

Activity 3

WriteaCprogramtoprintthefollowingpatternusingiterationcontrolstructure

```
kamshiga@kamshiga-VirtualBox: ~
 GNU nano 7.2                                   ta3.c
#include <stdio.h>
int main(){
 int i,j;

 for(i=1;i<=5;i++){
  for(j=1;j<=i;j++){
   printf("%d",j);
  }
  printf("\n");
 }

 return 0;

}
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ta3.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta3 ta3.c
kamshiga@kamshiga-VirtualBox:~$ ./ta3
1
12
123
1234
12345
kamshiga@kamshiga-VirtualBox:~$
```

Activity 4

Write a C program to do the following.

a) Create an array called data to store the monthly average rain fall for a period of one year.

b) Input the values from the key board and store them in the array.

c) Find the month with the maximum rainfall

d)Find the month with the minimum rainfall

```
kamshiga@kamshiga-VirtualBox: ~
  GNU nano 7.2                                          ta4.c
#include <stdio.h>
int main(){
 float data[12];
 int i;
 int maxMonth=0,minMonth=0;

 printf("Enter monthly average rainfall for 12 months:\n");
 for(i=0;i<12;i++){
  printf("Month %d: ",i+1);
  scanf("%f", &data[i]);
 }

 for(i=1;i<12;i++){
  if(data[i] >data[maxMonth]){
   maxMonth=i;
  }
  if(data[i] <data[minMonth]){
   minMonth=i;
  }
 }

 printf("\nMaximum rainfall is %.2f in Month %d ", data[maxMonth],maxMonth+1);
 printf("\nMinimum rainfall is %.2f in Month %d ", data[minMonth],minMonth+1);

 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ta4.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta4 ta4.c
kamshiga@kamshiga-VirtualBox:~$ ./ta4
Enter monthly average rainfall for 12 months:
Month 1: 2.0
Month 2: 3.0
Month 3: 56.09
Month 4: 67.0
Month 5: 78.90
Month 6: 99.0
Month 7: 45.67
Month 8: 34.00
Month 9: 23.0
Month 10: 20.09
Month 11: 34.00
Month 12: 55.67

Maximum rainfall is 99.00 in Month 6
Minimum rainfall is 2.00 in Month 1 kamshiga@kamshiga-VirtualBox:~$
```

Activity 5

Write a C program with a function named printNumbers() to print the numbers from 1 to 10

```
  GNU nano 7.2                                          ta5.c
#include <stdio.h>

void printNumbers(){
 int i;
 for(i=1;i<=10;i++){
  printf("%d",i);
 }
}

int main(){
 printNumbers();
 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano ta5.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta5 ta5.c
kamshiga@kamshiga-VirtualBox:~$ ./ta5
12345678910kamshiga@kamshiga-VirtualBox:~$
```

Activity 6

Write a C program with functions to find the grade of the students.

• Implement the void getmarks() function to input three marks. The marks must be read through the keyboard.

• Implement double calcAvg(double mark1, double mark2, double mark3) function with necessary parameters to calculate the average mark. 3

• Implement void findGrade(double avg) function with suitable parameters to gener ate the grade of the student based on the average mark. Use the criteria given in the below table to decide the grade

```
 GNU nano 7.2                                          ta6.c
#include <stdio.h>

double m1,m2,m3;

void getmarks(){
 printf("Enter mark 1: ");
 scanf("%lf",&m1);
 printf("Enter mark 2: ");
 scanf("%lf",&m2);
 printf("Enter mark 3: ");
 scanf("%lf",&m3);
}

double  calcAvg(){
 return (m1+m2+m3)/3;
}

void findGrade(double avg){
 if(avg >=75)
  printf("Grade A");
 else if(avg >=65)
  printf("Grade B");
 else if(avg >=55)
  printf("Grade C");
 else if(avg >=45)
  printf("Grade S");
 else
  printf("Grade F");
```

```
}

int main(){
 double avg;
 getmarks();
 avg=calcAvg();
 findGrade(avg);
 return 0;
}
```

```
kamshiga@kamshiga-VirtualBox: ~                    kamshiga@kamshig
kamshiga@kamshiga-VirtualBox:~$ nano ta6.c
kamshiga@kamshiga-VirtualBox:~$ gcc -o ta6 ta6.c
kamshiga@kamshiga-VirtualBox:~$ ./ta6
Enter mark 1: 80
Enter mark 2: 95
Enter mark 3: 88
Grade Akamshiga@kamshiga-VirtualBox:~$ ./ta6
Enter mark 1: 79
Enter mark 2: 60
Enter mark 3: 30
Grade Ckamshiga@kamshiga-VirtualBox:~$
```